

■ 1. Label Studio Shortcuts & Navigation

Key	Label / Action	Bounding Box Extent	Control	Function
1	yawning	Mouth region only	Ctrl + Enter	Submit & Next Frame
2	hand_over_mouth	Full head / face	Mouse Wheel	Zoom Canvas In / Out
3	drinking	Face + bottle together	Ctrl + Drag	Pan Canvas while Zoomed
4	phone_use	Hand + phone at ear	Delete / Bksp	Delete Selected Box
—	Negative Frame	0 Boxes (Alert / No Cue)	Ctrl + Z	Undo Last Edit

■ 2. Target Warning Cues & Bounding Box Rules (4 Frozen Cues)

Cue & Key	Visual Definition	Bounding Box Extent	Strict Exclusions (DO NOT ANNOTATE)
[1] yawning	Visibly active yawn distension / deep inhalation posture.	Mouth region only. Tight rectangle around lips/mouth.	• Talking/singing open mouth is NOT yawning. • Do NOT enclose the entire face/head.
[2] hand_over_mouth	Hand/fingers visibly touch, cover, or occlude mouth.	Full head and face. Enclose entire head + occluding hand.	• Hand resting on chin without covering mouth aperture is NOT hand over mouth. • Hand on wheel is NOT hand over mouth.
[3] drinking	Active drinking with bottle, cup, or can at mouth/face.	Face + bottle together. Enclose face and beverage container.	• Passive bottles in cup holders are NOT drinking. • Must depict active consumption posture.
[4] phone_use	Active handheld phone call (holding phone to ear/head).	Hand + phone at ear. Enclose phone, hand, and adjacent ear.	• Texting / lap browsing is EXCLUDED. • Hands-free voice calls are NOT phone use. • Mounted phones are NOT phone use.

■ 3. Protocol Rules & Edge Case Decision Matrix

Scenario / Ambiguity	Correct Action	Authoritative Benchmark Rationale
Driver looking at side/rearview mirrors or head rotated away	0 Boxes (Negative)	Drivers routinely perform mirror checks. In static 1-FPS split frames, there is no objective boundary between safe glances and real inattention; excluded to eliminate subjective label noise.
Driver yawning while hand visibly covers mouth	Both Boxes	Annotate hand_over_mouth (Full Head) AND yawning (Mouth). Overlapping boxes expected.
Driver holding phone to ear during call	phone_use	Annotate phone_use around hand + phone at ear (calling sense only; texting excluded).
Driver drinking water/beverage from bottle at lips	drinking	Annotate drinking around face + bottle together (active consumption).
Closed eyes / blinking during normal driving	0 Boxes (Negative)	eyes_closed is removed entirely from the single-frame benchmark ontology.
Single Static Frame Isolation Principle	Current Frame Only	Judge strictly by visible pixels in the current image. Do not infer blinks/motion from video memory.